In [17]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [18]:
```
data = {
    'StudentID': [1,2,3,4,5,6,7,8,9,10],
    'MathScore': [88, 92, np.nan, 70, 65, 85, 90, 87, 100, 55],
    'EnglishScore': [78, 85, 90, 68, 60, 82, 75, 88, 80, 94],
    'ScienceScore': [85, 90, 92, np.nan, 66, 88, 80, 95, 99, 60],
    'StudyHours': [10, 12, 15, 8, 6, 11, 9, 14, 13, 20],
    'Attendance': [95, 98, 100, 85, 80, 92, 90, 99, 97, 105]
}
```

In [19]:
```
df = pd.DataFrame(data)

print("Original Data:\n", df.head())
```

```
Original Data:
    StudentID  MathScore  EnglishScore  ScienceScore  StudyHours  Attendance
0          1       88.0            78          85.0          10          95
1          2       92.0            85          90.0          12          98
2          3        NaN            90          92.0          15         100
3          4       70.0            68           NaN           8          85
4          5       65.0            60          66.0           6          80
```

In [20]:
```
print("\nMissing values:\n", df.isnull().sum())
df['MathScore'] = df['MathScore'].fillna(df['MathScore'].mean())
df['ScienceScore'] = df['ScienceScore'].fillna(df['ScienceScore'].mean())
```

```
Missing values:
 StudentID       0
MathScore       1
EnglishScore    0
ScienceScore    1
StudyHours      0
Attendance      0
dtype: int64
```

In [21]:
```
df['Attendance'] = df['Attendance'].clip(0,100)
print("\nAfter cleaning:\n", df)
```

```
After cleaning:
    StudentID   MathScore  EnglishScore  ScienceScore  StudyHours  Attendance
0          1   88.000000            78     85.000000          10          95
1          2   92.000000            85     90.000000          12          98
2          3   81.333333            90     92.000000          15         100
3          4   70.000000            68     83.888889           8          85
4          5   65.000000            60     66.000000           6          80
5          6   85.000000            82     88.000000          11          92
6          7   90.000000            75     80.000000           9          90
7          8   87.000000            88     95.000000          14          99
8          9  100.000000            80     99.000000          13          97
9         10   55.000000            94     60.000000          20         100
```

In [22]:
```
outliers = df[df['StudyHours'] > 15]
print("\nOutliers in StudyHours:\n", outliers)
df['StudyHours'] = np.where(df['StudyHours'] > 15, 15, df['StudyHours'])
```

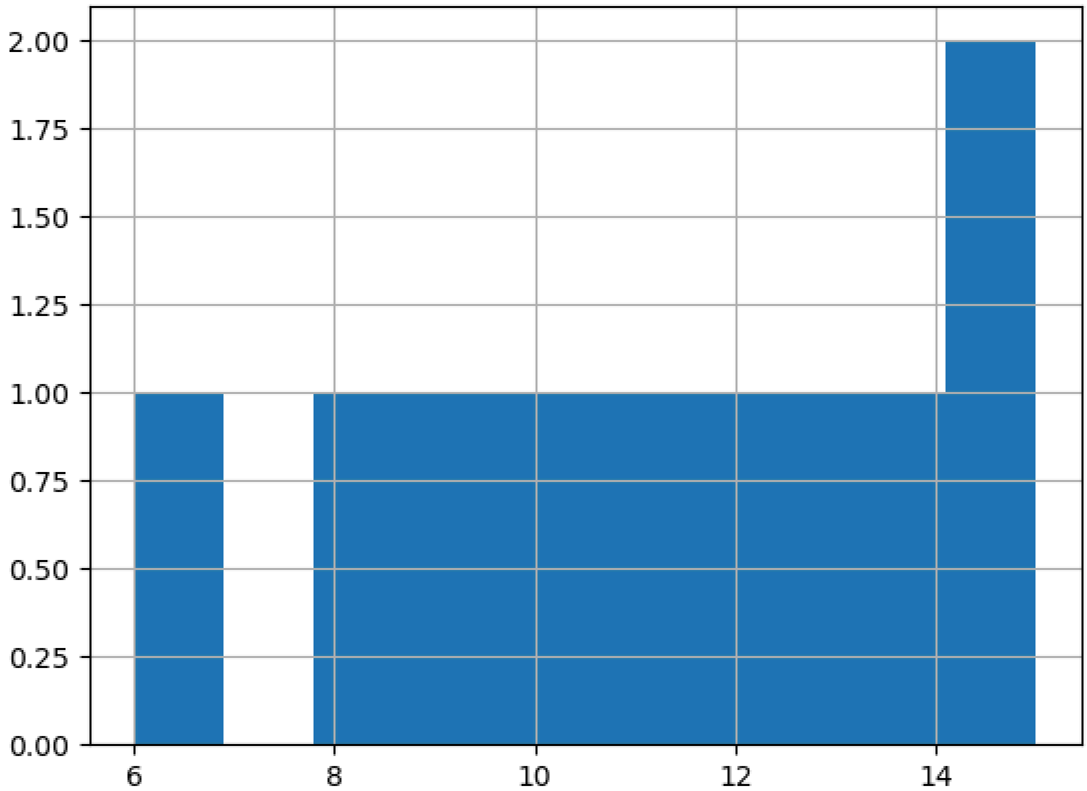
```
Outliers in StudyHours:
    StudentID  MathScore  EnglishScore  ScienceScore  StudyHours  Attendance
9         10       55.0            94          60.0          20         100
```

In [23]:
```
df['StudyHours_log'] = np.log1p(df['StudyHours'])
print("\nFinal Data:\n", df)
```

```
Final Data:
    StudentID   MathScore  EnglishScore  ScienceScore  StudyHours  Attendance  \
0          1   88.000000            78     85.000000          10          95
1          2   92.000000            85     90.000000          12          98
2          3   81.333333            90     92.000000          15         100
3          4   70.000000            68     83.888889           8          85
4          5   65.000000            60     66.000000           6          80
5          6   85.000000            82     88.000000          11          92
6          7   90.000000            75     80.000000           9          90
7          8   87.000000            88     95.000000          14          99
8          9  100.000000            80     99.000000          13          97
9         10   55.000000            94     60.000000          15         100

   StudyHours_log
0        2.397895
1        2.564949
2        2.772589
3        2.197225
4        1.945910
5        2.484907
6        2.302585
7        2.708050
8        2.639057
9        2.772589
```

In [24]:
```
df['StudyHours'].hist()
plt.show()
```



In [25]:
```
df.to_csv("academic_performance.csv", index=False)
```